



Olivine

Sea green

Main ingredients Silicon, oxygen, iron, magnesium

Crystal type Orthorhombic

Hardness on 1–10 scale 6.5–7

This is the main ingredient of the basalt bedrock of the ocean floor, and the peridotite that forms the upper mantle below Earth's crust. Large, well-formed

crystals are rare, and are highly valued as a gemstone known as peridot.



Fluorite

Eerie glow

Main ingredients Calcium, fluorine

Crystal type Cubic

Hardness on 1–10 scale 4

This simple chemical compound often forms twinned crystals that are like interlocking cubes. They are commonly green or purple. Some varieties of fluorite glow under ultraviolet light—a property known as fluorescence.



Rutile

Titanium ore

Main ingredients Titanium, oxygen

Crystal type Tetragonal

Hardness on 1–10 scale 6–6.5

Rutile is about 60 percent titanium—a light but strong metal that is used for many high-tech purposes including spacecraft. The crystals are often in the shape of long prisms, but also form clusters of slender needles.

Calcite

Soft and soluble

Main ingredients Calcium, carbon, oxygen

Crystal type Trigonal

Hardness on 1–10 scale 3

This common mineral can look like quartz, but it is much softer. Calcite often forms limestones that are gradually dissolved by slightly acid rainwater. The crystals grow in a variety of shapes.

Pyrite

Fool's gold

Main ingredients Iron, sulfur

Crystal type Cubic

Hardness on 1–10 scale 6–6.5

Very common and widespread, pyrite crystals have a glittering golden luster that has fooled many gold prospectors into thinking they have struck it rich. If pyrite is scratched with steel, it gives off sparks.



Tourmaline

Multicolored complexity

Main ingredients Silicon, oxygen, boron, plus various metals

Crystal type Trigonal

Hardness on 1–10 scale 7–7.5

Tourmaline is a complex compound of many elements. It forms long crystals that can be almost any color, and can even be multicolored.



Garnet

Historic gem

Main ingredients Silicon, oxygen, plus two or more metals

Crystal type Cubic

Hardness on 1–10 scale 6.5–7.5

Garnet crystals often occur in rocks that have been altered by intense heat and pressure, and can contain a variety of metals, which give them different colors. Pyrope, a magnesium-rich, blood-red variety, is a prized gem.